

Rahnuma

रहनुमा رہنما

A working guide to the ILM, PANINI and Zistgah stack, from bytes and scripts to compilers, Pāṇini and quantum substrates

प्रकृति

	written	Devanagari hub	Romenagri	and back
Devanagari	प्रकृति	→ प्रकृति	→ prak_ri_ti	→ प्रकृति
Telugu	ప్రకృతి	→ प्रकृति	→ prak_ri_ti	→ ప్రకృతి
Bengali	প্রকৃতি	→ प्रकृति	→ prak_ri_ti	→ the same identity
Urdu, as usually written	ہندوی	→ हनदवी	→ _hana_dawee	→ हनदवी
Urdu, with zer and sukun	ہندوی	→ हिन्दवी	→ _hin_dawee	→ हिन्दवी

One identity, many scripts. Telugu and Bengali meet Devanagari in one hub, become the same Romenagri word, and come back exactly. Urdu meets the hub at what is written: without its short-vowel marks the word reads hanadawee; with zer and sukun it is hindawi. Every cell here was produced by the Hindawi tools while this guide was built.

Abhishek Choudhary, AyeAI. Version 2.0.0, 3 October 2026.

Rahnuma, the edition for the field: Agriculture, forestry, fisheries and veterinary

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Part I: Orientation

1. How to use this guide

In plain words. This guide explains a large body of work on computers, languages and intelligence. You can read it, run the programs on your own computer, or ask an AI to help you, and every example in it was really run.

This guide is for anyone who wants to use, check or extend the work gathered in the *zistgah*, *project-ilm* and *hindawiai* organisations on GitHub: students preparing for GATE, engineers who write compilers, firmware or chip layouts, linguists who work with scripts and sound, and researchers in quantum computing and AI. It asks for curiosity and a laptop, and nothing else.

There are four ways in.

- **Understand.** Part II gives the whole architecture first: where the work comes from, how the parts compose, each term with its nearest analogue and the difference, and the three Acts.
- **Read.** Parts IV and V explain the ideas the work rests on, with the mathematics written out: how numbers and letters sit in memory, what makes a transliteration reversible, how Pāṇini's grammar computes, how a compiler turns text into machine instructions, and how an oscillator network and a quantum circuit attack the same problem.
- **Run.** Part III gives the exact commands to set up Linux, Windows with WSL, or macOS, and to run every component of the estate on your own machine.
- **Ask.** Chapter 24 shows how to put any AI assistant to work on these repositories, and how to check what it tells you.

Conventions used throughout:

- Every example that could run while this guide was being built did run, and the text shows its real output, marked *Executed while this guide was built*. The package's own `gate`, `bash ops/verify.sh`, runs those examples again and fails if any output differs from what is printed here.
- Status words are used exactly. **Tested:** a test in the repository asserts it. **Executed:** it ran and its output is shown. **Written:** the code exists but has not been run here. **Mocked:** a stand-in replaces hardware that is not attached. **Planned:** described, not built.
- Everything said about the repositories was retrieved from the repositories themselves on 29 September 2026: their listings on GitHub, their `CITATION.cff` files and their READMEs. Where a repository does not record something, this guide says so instead of filling the gap.
- A dagger, †, marks a statement that follows the author's working notes and write-ups as supplied on 29 September and 3 October 2026, rather than the repositories.
- Sanskrit is transliterated in IAST. The spelling is British.

The work lives at github.com/zistgah, github.com/project-ilm and github.com/hindawiai, on the site zistgah.org, and in the Zenodo records listed in Chapter 12.

3. Start here: find your way in

In plain words. Four steps to find your part of the guide: who you are, your board or field, what you want to do, and your path with a PDF made for you.

This guide serves a school student and a senior researcher; a village health worker and an investor. Nobody needs all of it. Four steps find your part, on the site or with this page:

1. **Who you are.** A school student, a college student, a postgraduate or doctoral researcher; a teacher; an academic researcher; a working professional, someone in public service, policy or law, someone in community, NGO or village work, an artist, writer or maker; an industry partner or investor; or an AI agent working for a person.
2. **Your board or your field.** A school student chooses a board, ICSE, ISC, CBSE, IGCSE, IB, a state board or another; and a class. Everyone else chooses a field from the international standard classification of fields of study, ISCED-F 2013, the same one FAKIR's lattice uses. It places each subject where its own scholars would: political science is under social and behavioural sciences, not the humanities; law is under business, administration and law; linguistics is under languages.
3. **What you want to do.** Learn the basics, run the software, study or research, teach, contribute, or adopt, fund or invest.
4. **Your path and your edition.** The site lists your chapters in order, with reading times and the components to try first, and offers a PDF edition made for you. Every edition below is a printed selection from this one guide, so all of them stay in step.

Edition	PDF
School student	role-school.pdf
College student	role-college.pdf
Postgraduate or doctoral researcher	role-pg.pdf
Teacher or lecturer	role-teacher.pdf
Academic researcher	role-researcher.pdf
Working professional or engineer	role-professional.pdf
Public service, policy or law	role-public.pdf
Community, NGO or village work	role-community.pdf
Artist, writer or maker	role-creator.pdf
Industry partner, investor or funder	role-industry.pdf
Generic programmes and qualifications	field-00.pdf
Education	field-01.pdf
Arts and humanities	field-02.pdf
Social sciences, journalism and information	field-03.pdf

Edition	PDF
Business, administration and law	field-04.pdf
Natural sciences, mathematics and statistics	field-05.pdf
Information and Communication Technologies (ICTs)	field-06.pdf
Engineering, manufacturing and construction	field-07.pdf
Agriculture, forestry, fisheries and veterinary	field-08.pdf
Health and welfare	field-09.pdf
Services	field-10.pdf

The first public version of this guide, written for advanced researchers and senior professionals, stays as it was at [version 1](#).

Part II: One architecture

10. Three families, set apart: the AyeAI Triad, GramSheel and Project VIKRAM, Humanesque and its projections

In plain words. Three groups explained one at a time: the AyeAI Triad of thinking, body and communication; GramSheel and Project VIKRAM for villages; and Humanesque with the names it takes in each culture.

Three families of work are easy to blur into each other or into the rest of the estate. This chapter sets each apart, with what its own record says.

The AyeAI Triad

The Triad has three members, AyeAI®, AyeAM® and AyeCNSe®, with AyeAI at the apex and AyeCNSe and AyeAM at the base. It is a triad, not a pipeline and not a loop: a loop running from the environment through AyeCNSe, AyeAI and AyeAM back to the environment was proposed elsewhere and rejected. Its formal closure gives each member a tuple:

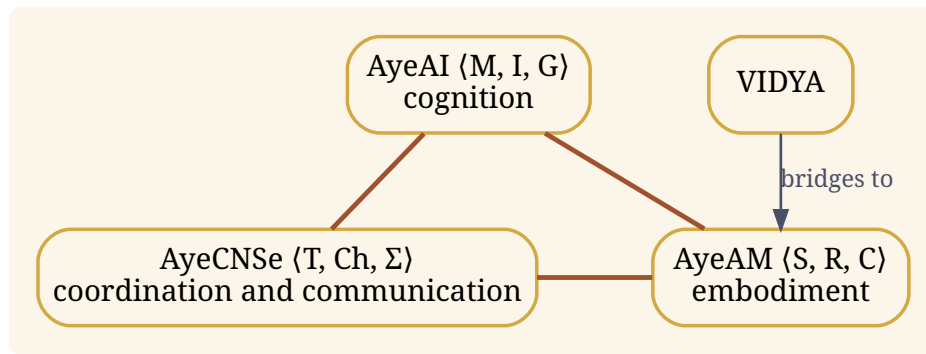


Figure 8. The AyeAI Triad: a triad with AyeAI at the apex, not a loop and not a pipeline.

$$\mathbf{AyeAM} = \langle S, R, C \rangle, \quad \mathbf{AyeAI} = \langle M, I, G \rangle, \quad \mathbf{AyeCNSe} = \langle T, Ch, \Sigma \rangle.$$

AyeAI is cognition, the apex: “the inclusive AI”, in the words of its site, and the organisation [ayelai](#) holds its platforms, among them an open public stack for synthetic intelligence (opssi), virtual deployment infrastructure for scientific and cognitive computing (ayevdi), Staggered Upload™ (upload), the AyeAI Singularity Public License (spl) and the NI2A2 work.

AyeAM is embodiment: the body through which intelligence acts, across ground, air, water, space and habitat. VIDYA bridges to AyeAM without being a member of the Triad.

AyeCNSe is coordination and communication across media: the form studio at [forms.ayecnse.site](#) and the Human Context Model paper at [hcm.ayecnse.site](#) are its published faces.

Two repositories mark themselves as PEDLER and AyeAI Triad work: PRATIK’s kernel and Zamin’s ternary silicon.

GramSheel and Project VIKRAM

The GramSheel Foundation is the social architecture of the estate and the older of its lines: † it began as the Nature Welfare Society in 1993. Its principles, HEJLP, are health, education, justice, livelihood and peace. Its Village Knowledge Center (Universal Device) was a Stockholm Challenge finalist in 2010, and TWISHA runs under the same foundation.

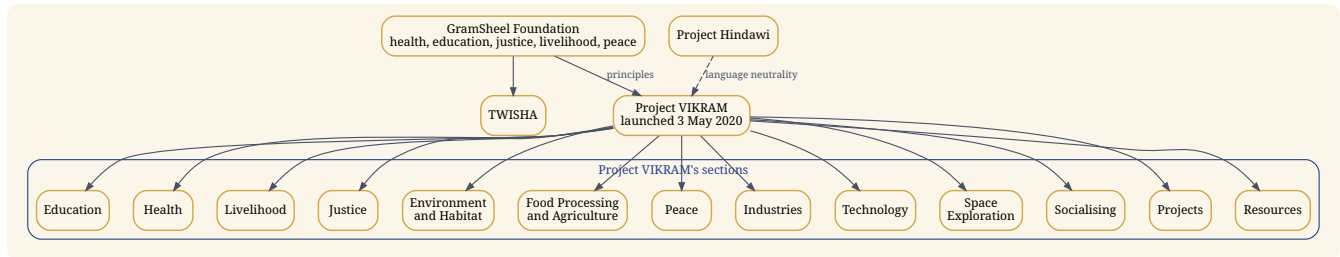


Figure 9. GramSheel and Project VIKRAM, with VIKRAM's sections as its own pages name them.

Project VIKRAM, Virtualized Infrastructure for Knowledge-driven Rural Ascension Management, is † the technical implementation of GramSheel's principles. It was launched on 3 May 2020 as the Project Vikram Journal, “an inclusive rural digitalization platform inspired by the principles of GramSheel”, under the motto गाँव बढ़ेंगे तो सब बढ़ेंगे, when the villages grow, everyone grows. Its own pages (pvjournal.github.io) name its components:

Component	What its page covers
Education	learning and schooling
Health	healthcare, open to submissions as issues, with tools for an accessible virtual hospital
Livelihood	the ISIC domains of economic activity
Justice	social justice
Environment and Habitat	the living environment
Food Processing and Agriculture	agriculture and food
Peace	harmony through social dialogue
Industries, Technology and Space Exploration	the technical frontiers it reaches towards
Socialising, Projects and Resources	the community and its shared material

With Project Hindawi, its page says, VIKRAM works towards language neutrality across technical fields. In the three Acts, GramSheel and Project VIKRAM stand on the divider between Acts II and III.

Gram Swaraj and the questions behind GramSheel and VIKRAM

Gandhi's *Hind Swaraj* (1909) argued that self-rule is more than a transfer of authority, and his *Constructive Programme: Its Meaning and Place* (1941) placed the rebuilding of daily life inside the work of independence: its eighteen items include village industries, village sanitation, basic and adult education, the provincial languages and a national language, economic equality, the kisans, labour, the adivasis and women. The estate does not present itself as a restatement of these ideas. The point of contact is a question that becomes unavoidable once computation is infrastructure: can a community govern itself if the knowledge and the machines its decisions depend on are controlled elsewhere?

On Gandhi Jayanti, 2 October 2026, the author put the questions behind GramSheel, Project VIKRAM and Humanesque in his own words. Among them:

- Can rural communities become producers of knowledge and technology, not merely consumers of systems designed elsewhere?
- Can technology distribute capability without concentrating power?
- Can artificial intelligence strengthen individual and community sovereignty rather than replace it?
- Can knowledge remain plural, multilingual and culturally grounded while becoming computationally executable?
- Can verification, provenance and human agency become constitutional properties of intelligent systems?
- Can GramSheel build the local knowledge infrastructure, can Project VIKRAM turn that infrastructure into practical capability, and can Humanesque provide a larger architecture in which intelligence, identity, agency, provenance and equity remain connected?

His answer to the old division of the world is a direction rather than a side: to march together, not as a Global South or a Global North, but towards a Global Centre, where capability is distributed and humanity remains sovereign.

Humanesque and its projections, with their names

Humanesque is the core, with its site at humanesque.site and its organisation at hmnsq: the shared executable typed hypergraph, the ontology federation, the common execution and evidence spine, and the components every projection reuses. Its constitutional ground is PoIE, from which sovereignty recurses to every level. Its core stem repositories all belong to Humanesque; each projection gives them its own culturally sensitive names and sites.

Kaivalyik, the Indic projection, including the Advaita framing (kaivalyik.org, organisation [kaivalyik](https://kaivalyik.org)): consciousness, intelligence and transcendence; it carries Synthemetic Pragmatic Realism, Kaivalyik AGI and Jyotish. **Zistgah**, زیستگاه, the Persian and Islamicate projection (zistgah.org, organisation [zistgah](https://zistgah.org)): life, planet and ecology, a habitat to live and work in; it carries the elements, the dome and the habitats on Earth (Asli Zistgah), the Moon (Zistgah-e-Mahtab) and Mars (Zistgah-e-Bahram, † conceived earlier as Mangal Kamna). **Cosmopolis**, the Greek, Western and classical projection (cosmopolis.org, organisation c-polis): societies, civilisations, space and deep time.

The stem names below are the projection-neutral names of the components. Today the working repositories live in the zistgah organisation, so most of them carry their Zistgah names; two stems already show a second projection, HPS in Hindawi and CHAKRA in Jyotish.

Stem, in Humanesque	Zistgah	Kaivalyik
The programming system	Urdu edition (urdu-ilm)	Hindawi
The observatory and its calendars	Chakra, the turning sky	Jyotish
The ground: balanced-ternary silicon	Zamin, زمین	
The water: the factory process	AAB, آب	
The air: the environmental replica	Fiza, فضا	
The shared virtual dome	the dome	
The Cognitive Workbench	Mez, میز	
The press	Matba, مطبع	
The cutting room, the visual cyler	Khwab, خواب	
The listening room, the audio cyler	Awaz, آواز	
The immersive cyler	Tilasm, طلسم	
The embodied cyler	Pench	
The record	Yadein, یادیں	
The book template	Kitab, کتاب	
The domain spine	Dhancha, ڈھانچہ	
Communication engineering	Ertabat, ارتباط	
Device control	Zasab, اعصاب	
The door for any AI	Darwaza, دروازہ	
Scan and store	Tasvir, تصویر	
The plates	Lawh, لوح	
The parser federation	Tajziya, تجزیہ	
The guide	Rahnuma, رہنما	
The Geometry of Becoming	Tabdili, تبدیلی	
The Geometry of Living	Zindagi, زندگی	
The Geometry of Persisting	Qaem, قائم	
Topological Bildung	Tarbiyat, تربیت	

Some stems keep one name in every projection: Humanesque, ILM, Romenagri, PANINI with PANINIq, PANINIb and PANINIphy, PEDLER, QEDLER, PRATIK, CEM, UKOP, FAKIR, GENIE, TransEg, PoIE, VGC, COPA, and the provenance systems Tok DOI, spiguard, Candor and Misty DOI. Kaivalyik's and Cosmopolis's own names for the rest are given with their organisations, as each projection takes its repositories.

Part V: Working with the estate

29. The roadmap, for every collaborator and every domain

In plain words. The plan to finish the releases and open the work to everyone: what happens when, which lane each kind of person can join, and every field of work, from farming to medicine to software.

The plan to complete the releases and to open the estate to collaborators is kept as data and run as a GitHub project: one board, an epic for each phase, and an issue for every piece of work, each with its finding, its task, the exact gate that proves it done, the evidence to attach, and what it depends on. The same data gives every collaborator a lane and every domain a lane, so anyone can find where they fit and what is open to them. The roadmap page filters it all by phase, by collaborator lane, by domain family and by owner; at zistgah.org/roadmap.

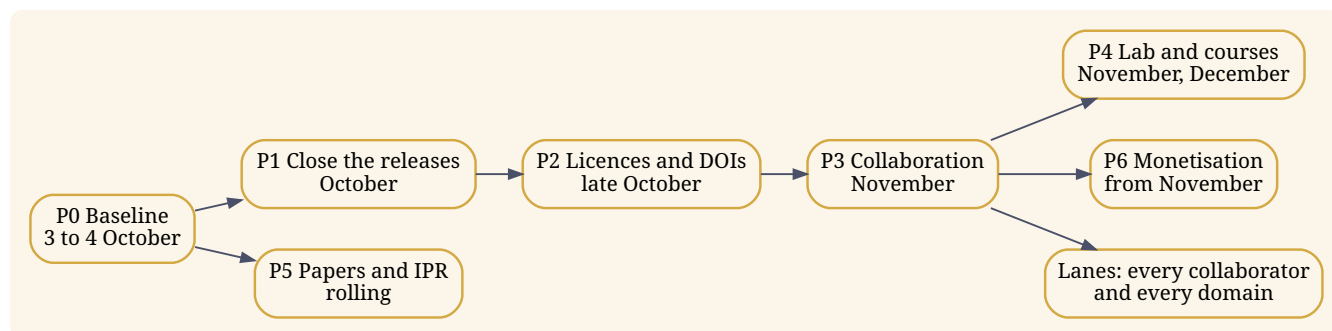


Figure 12. The plan's phases; the lanes run alongside from P3.

Phases

Phase	Window	Goal	Issues
P0 Baseline	Weekend of 3 to 4 October 2026	One measured inventory, the board, tokens with the right scopes, and the list of rulings.	4
P1 Close the releases	10 to 25 October 2026	Everything built is released, and every defect found by execution is fixed.	19
P2 Licences and DOIs	24 October to 8 November 2026	Every repository carries a licence file and correct metadata; minting covers the estate by family.	5
P3 Collaboration infrastructure	31 October to 15 November 2026	A newcomer can go from Rahnuma to a merged pull request in one sitting.	7
P4 Reference lab and courses	November to December 2026	The reference lab runs from launch scripts, with GATE-aligned modules and a pilot.	5

Phase	Window	Goal	Issues
P5 Papers and IPR	Rolling, from October 2026	Disclosure ledger first, then preprints of the four papers with results.	7
P6 Monetisation	From November 2026	Two offers defined and priced, on the dossier standard.	3
Lanes: every collaborator and every domain	Continuous, from October 2026	Every collaborator type and every domain family has a visible lane on the board, with an entry point, open work and a way to propose more.	62

Collaborator lanes

Lane	Start here	Contribute	Issues
School students, 10 to 17	the Rahnuma view for school students; CHAKRA, PANINI's studio, GENIE and the FAKIR dome	learning paths, translations of plain-words summaries, questions	1
Undergraduates: BE, BTech, BSc, BS	the GATE view; Running every component	good first issues, component fixes, capstones seeded from FAKIR	52
Postgraduates and doctoral researchers	the researcher views; Appendices C and D	theses and papers on the open questions, measured results, reviews	52
Teachers, lecturers and lab builders	the teacher view; the reference lab	course modules, lab sheets, pilots, reviews of plain-words text	9
Software engineers	the engineer view; the compilers chapter	code, tests, gates, tooling, pull requests	22
Hardware, embedded and chip engineers	Below the software; zamin, pratik_core_mvp, jugaad28	SPICE, layouts, firmware, device control, ESP32 work	8
Linguists, translators and language workers	the linguist view; writing systems and Pāṇini	language tables, corpora, transliteration rules, measurements	7
Researchers in AI, quantum and cognition	the researcher view; PANINIQ; PEDLER	models, experiments, validation, papers	7
Researchers in the humanities and social sciences	the humanities view; Appendix C	studies of agency, language, knowledge and community	4
Clinicians, health workers and medical researchers	VIKRAM's health section; CEMb and QNBCI	clinical questions, partnerships, review, health-domain tasks	3

Lane	Start here	Contribute	Issues
Community workers, NGOs and village institutions	GramSheel and Project VIKRAM	local needs, pilots, language and field knowledge	30
Farmers, artisans and local enterprises	FAKIR's domain lanes; VIKRAM's livelihood and agriculture sections	domain knowledge, problems worth solving, field testing	24
Artists, writers, filmmakers and musicians	the maker view; the cyclers	works made with the cyclers, recorded as method	3
Makers, roboticists and manufacturers	PANINIphy, pench, transeg	physical realizations, parts, embodiment	6
Policy makers, administrators and public bodies	Appendices C and E; sovereignty in Chapter 19	policy questions, adoption, public infrastructure	5
Lawyers, IPR and standards professionals	provenance and licences; the disclosure ledger	licence review, IPR clearance, standards work	14
Industry partners and companies	Appendix E's adoption path; the component cards	integration, pilots, paid offers, support	9
Funders, sponsors and investors	this roadmap and its measures	funding of lanes, labs and pilots	4
Institutions: colleges, universities and labs	the reference lab and the GATE modules	pilots, courses, joint research	13
AI agents working under a human	any issue's URL; Chapter 22, how to use any AI	issue work through pull requests, with gate output attached	65

Domain families

From FAKIR's lattice: [zistgah/fakir data/lattice.json](#) (ISIC Rev.4, ISCO-08, ISCED-F 2013), 1,505 nodes.

Family	Name	Domains within	Issues
ISIC A	Agriculture, forestry and fishing	54	1
ISIC B	Mining and quarrying	29	1
ISIC C	Manufacturing	232	4
ISIC D	Electricity, gas, steam and air conditioning supply	7	1
ISIC E	Water supply; sewerage, waste management and remediation activities	18	1
ISIC F	Construction	22	1
ISIC G	Wholesale and retail trade; repair of motor vehicles and motorcycles	66	1

Family	Name	Domains within	Issues
ISIC H	Transportation and storage	36	1
ISIC I	Accommodation and food service activities	15	1
ISIC J	Information and communication	42	22
ISIC K	Financial and insurance activities	31	1
ISIC L	Real estate activities	5	1
ISIC M	Professional, scientific and technical activities	35	11
ISIC N	Administrative and support service activities	51	1
ISIC O	Public administration and defence; compulsory social security	11	1
ISIC P	Education	14	7
ISIC Q	Human health and social work activities	21	1
ISIC R	Arts, entertainment and recreation	19	1
ISIC S	Other service activities	26	1
ISIC T	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	8	1
ISIC U	Activities of extraterritorial organizations and bodies	3	1
ISCO 1	Managers	46	1
ISCO 2	Professionals	125	1
ISCO 3	Technicians and Associate Professionals	109	1
ISCO 4	Clerical Support Workers	41	1
ISCO 5	Service and Sales Workers	57	1
ISCO 6	Skilled Agricultural, Forestry and Fishery Workers	30	1

Family	Name	Domains within	Issues
ISCO 7	Craft and Related Trades Workers	85	1
ISCO 8	Plant and Machine Operators, and Assemblers	57	1
ISCO 9	Elementary Occupations	50	1
ISCO 0	Armed Forces Occupations	9	1
ISCED 00	Generic programmes and qualifications	6	1
ISCED 01	Education	5	4
ISCED 02	Arts and humanities	13	5
ISCED 03	Social sciences, journalism and information	8	5
ISCED 04	Business, administration and law	10	7
ISCED 05	Natural sciences, mathematics and statistics	13	4
ISCED 06	Information and Communication Technologies (ICTs)	4	16
ISCED 07	Engineering, manufacturing and construction	15	4
ISCED 08	Agriculture, forestry, fisheries and veterinary	9	1
ISCED 09	Health and welfare	12	1
ISCED 10	Services	14	1

The AGI layers are L0 Hardware Definition, L1 Verification, L2 Synthesis/Compilation, L3 Firmware/Embedded, L4 Systems, L5 Parallel, L6 Distributed/Cloud, L7 AI/Deep Learning, L8 Robotics/Embodiment, L9 AGI/ASI. Every domain node, crossed with every layer and every language in ILM's registry, is a possible seed:

$$|S| = |D| \times |L| \times |\Lambda| = 1,505 \times 10 \times 7,867 = 118,398,350.$$

How a collaborator comes in

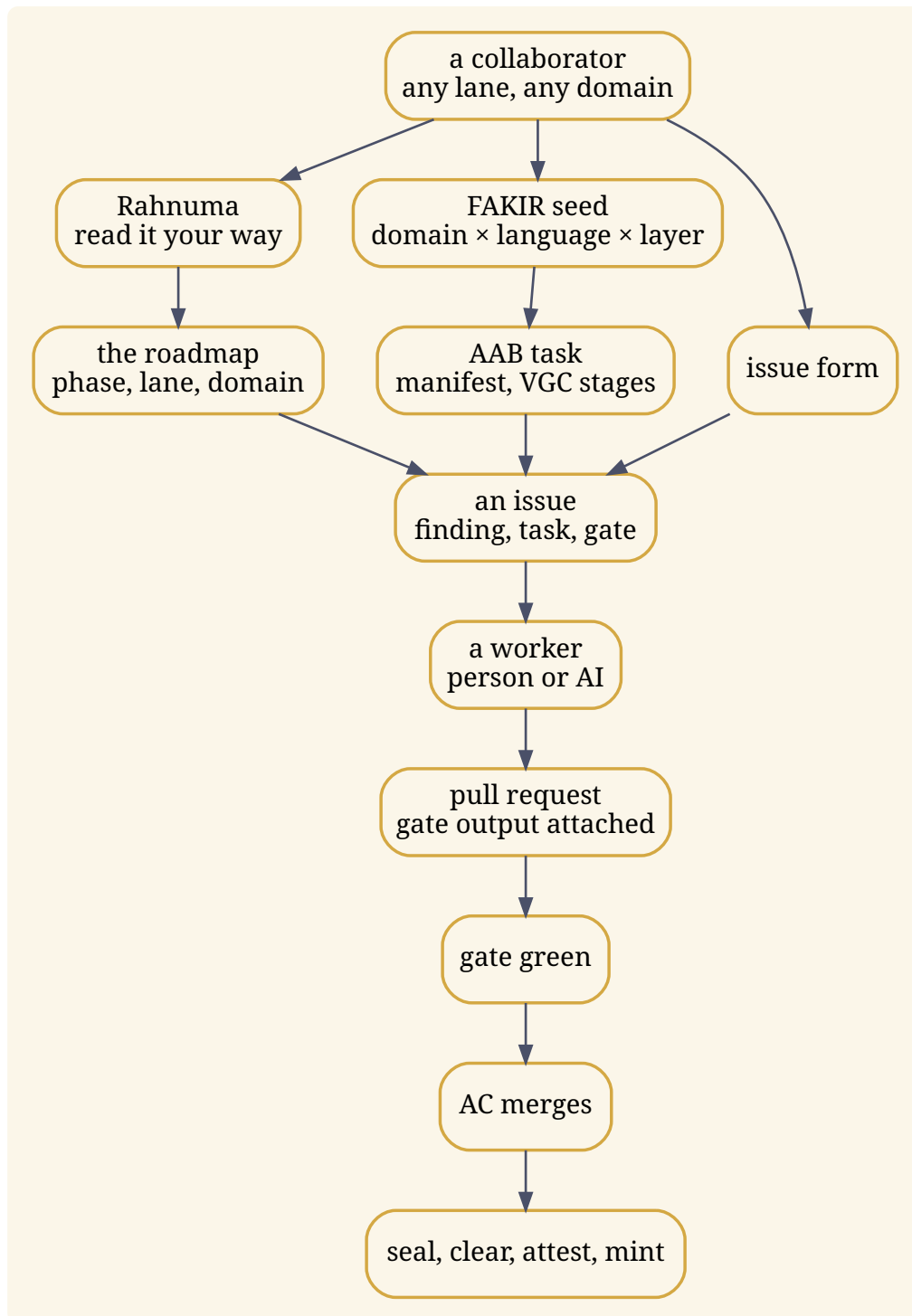


Figure 13. How any collaborator, in any domain, goes from the guide to a sealed contribution.

Four issue forms in [zistgah/governance](#) are the ways in: **Join as a collaborator**, with your lane, domains and languages; **Propose a domain task**, seeded the way FAKIR seeds it, with a domain code, an AGI layer, a language, an intent and a gate; **Report a finding**, with the commands run and what they printed; and **Request a ruling**, for the questions only the author can settle. Every issue says which owner class takes it: a Claude Code session, ChatGPT, Gemini or Fable as workers, or the author himself for push, Pages, mint, merges and rulings.

The work flows the same way for everyone. A worker, a person or an AI, opens a pull request with the gate's output attached; the author merges what is green; anything pushed or minted carries its Candor receipt and goes through seal, clear, attest and mint. Each Saturday the status digest shows every issue as closed, ready, or blocked and by what.

Appendices

E. For practitioners: teachers, policy makers, communities, makers and engineers

In plain words. A guide for teachers, policy makers, community workers, artists and engineers: where to start and how to take part.

Teachers and lab builders. Start from the reference lab, the GATE chapter and the component cards, which double as laboratory sheets: each says what to install, what to type and what a student should see (Chapters 16, 26 and 15). The reader views let each student follow a path suited to their age and background, and remember what they have read.

Policy makers and administrators. The questions that matter are in Chapter 10 and Appendix C: sovereignty in its testable technical sense, the ability to rebuild, verify, modify and run (Chapter 21); provenance that makes claims checkable; and infrastructure that leaves capability with communities. The executable contracts of the repositories show what verifiable governance of software looks like in practice.

Community workers and health practitioners. Project VIKRAM's sections cover education, health, livelihood, justice, environment and habitat, food and agriculture, and peace; its health section accepts submissions as issues. The Urdu edition and the Brahmi hub show work in people's own scripts; CHAKRA and Jyotish run offline.

Artists, writers and makers. The cyclers produce print, film, sound, immersive work, embodied work and the record; matba turns posters into a sealed book; GENIE writes and paints; PANINIphy turns a physical idea into parts. Every cycle is recorded as method, so a piece of work can be repeated, shared and credited.

Engineers in industry. Adopt one component at a time: read its card, clone it, run it, read its `CONTRACT.md` and its gate, check its DOI and provenance, then integrate it through its interface. † The notes give the full adoption ladder: the canonical artifact, a one-page explanation, the nearest industry analogue and the exact difference, a working demonstration, the repository, the DOI and timestamp, the interface, a case study, and the commercial route.

Contributors. Fork, open an issue, send a pull request. Every repository's gate runs on your change; every claim must point to its evidence; the AI you use is welcome, and its report that something passed is not the evidence that it did.

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